Mark “Xavier” Bonavita

xvr0612@gmail.com • (203) 903-6638

https://xvr6.dev/

Education

Worcester Polytechnic Institute (WPI) | Worcester, MA — **GPA 3.62 / 4.0**

Masters of Science in Computer Science — 08/2025 - Present

Bachelor of Science in Computer Science — 08/2022 - Present

Technical Skills

Languages	C/C++, Java/Kotlin, JavaScript/TypeScript, Elixir, JSON, Python, SQL, HTML/CSS
Tools/Platforms	Node.js, Flask, SQLite, PostgreSQL, Azure, Jupyter, Git, React
Relevant courses	Advanced Computer Networks, Mobile & Ubiquitous Computing Software Engineering, Operating Systems, Database Systems

Projects

AI-Powered Species Identification, *WPI MQP supported by NSF* | Worcester, MA — May 2025 - August 2025

- Developed an AI-powered species identification pipeline using melting curve analysis and a Vision Transformer (ViT) model to classify DNA denaturation patterns as images
- Achieved 95.35% classification accuracy with less than 0.02 loss by leveraging Optuna for hyperparameter tuning and synthetic data generation for class balance
- Built a cross-platform mobile app enabling real-time species identification in the field by interfacing with a remote classification server
- Enhanced wildlife enforcement capabilities by providing a scalable, accessible tool for rapid species identification, even from partial or processed biological samples
- Demonstrated the application of AI in conservation biology and forensic science to support biodiversity protection and combat illegal wildlife trade

Privacy in Home IoT Devices, *WPI IQP Project* | Worcester, MA — August 2024 - March 2025

- Collaborated in a team of four and a faculty advisor to investigate trends, challenges, and opportunities in the home Internet of Things (IoT) industry.
- Conducted comprehensive research on consumer behavior, privacy concerns, device interoperability, and market growth within the smart home ecosystem.
- Collected and analyzed quantitative data using statistical tools and a survey to evaluate usage patterns and security vulnerabilities across various IoT platforms.
- Authored a detailed project paper presenting key findings, supported by statistical analysis, and provided recommendations for enhancing user privacy and system integration in smart home environments.

furspace.blog, *Humanities Practicum* | Worcester, MA — January 2025 - March 2025

- Alongside two other artists to translate a shared creative vision into a collaborative art-driven website with rich storytelling.
- Designed and implemented a nostalgic early-2000s aesthetic, drawing inspiration from platforms like MySpace to evoke a distinct digital retro atmosphere.
- Developed a narrative-driven experience, integrating a continuous storyline throughout the blog-style layout to enhance visitor engagement and immersion.
- Contributed to the creative direction and visual identity, balancing unique artistic styles within a cohesive design framework.
- Developed custom scripts to automate the display of multimedia content and blog entries, ensuring consistent formatting.

When2Goat, *Meeting and Schedule Coordination Website* | Worcester, MA March 2024 - May 2024

- Worked in a team of four to develop a full-stack application within 5 weeks, following Agile/Scrum methodologies.
- Designed and implemented a scalable Flask (Python) backend with an Azure-hosted PostgreSQL database.
- Streamlined user experience (UX) by delivering a simple yet feature-rich interface that functions efficiently across various device types and screen sizes.
- Integrated WPIs Microsoft Single Sign-On (SSO) and Microsoft Entra API for secure identity verification and seamless Microsoft Calendar integration.

Neo, *Discord Application (commonly known as: Bot)* | Worcester, MA March 2019 - Present

- Developed a web server using Node.js to interact with Discord's API, enabling a responsive bot with a wide range of advanced functionalities.
- Utilized both hot and cold SQLite 3 databases to efficiently store and manage session data, user preferences, and server configurations.
- Embedded multiple external libraries while modularizing code into independently executable segments, ensuring scalability and ease of operation.